

# Lecture Notes on Differential Equations and Transform Techniques

*Spring 2025*

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

## References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

## EE1060: Differential Equations and Transform Techniques

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## Topics covered :

1. Symmetry and its impact on Fourier Coefficients
2. Effect of time-shift and Derivative on Fourier Coefficients

- **Example:** Consider a square wave as shown in Figure 1(a). The Fourier Series coefficients  $C_k$  for the square wave

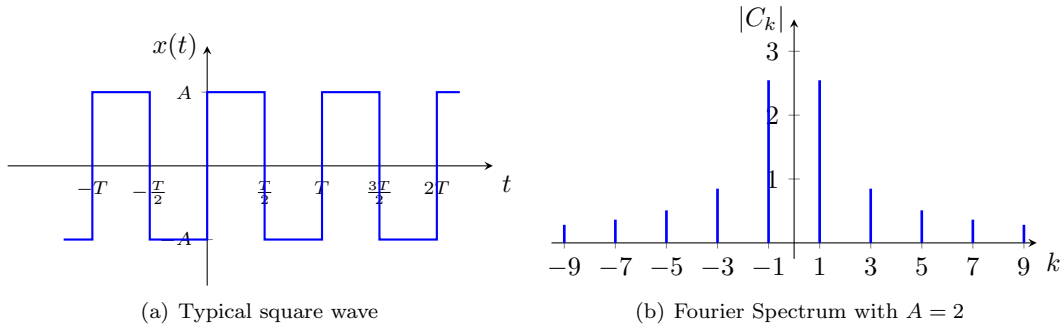


Fig. 1: Square wave and its Fourier Spectrum

are given by

$$\begin{aligned}
 C_k &= \frac{1}{T} \int_0^T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \left[ \int_0^{T/2} A e^{-jk\omega_0 t} dt - \int_{T/2}^T A e^{-jk\omega_0 t} dt \right] \\
 &= \frac{1}{T} \left[ \frac{A}{-jk\omega_0} \left( e^{-jk\omega_0 t} \Big|_0^{T/2} + e^{-jk\omega_0 t} \Big|_{T/2}^T \right) \right] \\
 &= \frac{jA}{2\pi k} (2e^{-jk\pi} - 2) \\
 &= \begin{cases} -\frac{2A}{k\pi} j & \text{when } k \text{ is odd} \\ 0 & \text{when } k \text{ is even} \end{cases}
 \end{aligned} \tag{1}$$

The square wave signal has odd harmonics and its magnitude of harmonic content falls over as  $\frac{1}{k}$ . Further, the coefficients of trigonometric Fourier series can be computed as (when  $k$  is odd)

$$a_k = C_k + C_{-k} = 0 \text{ and } b_k = j(C_k - C_{-k}) = \frac{4A}{k\pi} \tag{2}$$

- It is a common practice in signal analysis to examine the frequency content of a signal by plotting the relationship between frequency (or, equivalently, the bin index  $k$ ) and the magnitude of the Fourier coefficients  $C_k$ , as well as the frequency (or bin index  $k$ ) and the phase of the Fourier coefficients  $C_k$  (although the phase plot is less commonly used in basic analysis). A plot that depicts the frequency (or bin index  $k$ ) versus the magnitude of the Fourier coefficients  $C_k$  is commonly referred to as the **Fourier Spectrum**. The Fourier spectrum of the square wave shown in Fig. 1(a) is presented in Fig. 1(b).

- **Fourier series of an Even Periodic Function:** Let  $g(x)$  be an even periodic function (i.e.,  $g(-x) = g(x)$ ) with period  $L$ . The Complex Fourier coefficients of  $g(x)$  are given by

$$\begin{aligned}
 C_k &= \int_{-\frac{L}{2}}^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx = \frac{1}{L} \left[ \int_{-\frac{L}{2}}^0 g(x) e^{jk\omega_0 x} dx + \int_0^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx \right] \\
 &= \frac{1}{L} \left[ \int_{\frac{L}{2}}^0 g(-u) e^{jk\omega_0(-u)} (-du) + \int_0^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx \right] \\
 &= \frac{1}{L} \left[ \int_0^{\frac{L}{2}} g(u) e^{-jk\omega_0(u)} (du) + \int_0^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx \right] = A + A^* \text{ where } A = \frac{1}{L} \int_0^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx
 \end{aligned} \tag{3}$$

On the other hand, the trigonometric Fourier coefficients for an even periodic function are given by

$$a_k = (C_k + C_k^*) = 2(A + A^*) \text{ and } a_k = 2j(C_k - C_k^*) = 0 \tag{4}$$

Thus, the trigonometric Fourier series for an even periodic function contains only cosine terms in the synthesis. This is expected because the even symmetry of the function ensures that all sine components (which are odd functions) cancel out, leaving only the cosine components, which are even functions. Consequently, the Fourier series representation of an **even periodic function involves only cosine terms**. Similarly, it can be shown that the **Fourier series of odd periodic function** contains only sine terms.

- **Fourier series of functions with half-wave symmetry:** A periodic function  $g(x)$  with period  $L$  is half-wave symmetric if  $g(x + \frac{L}{2}) = -g(x)$ . The complex Fourier coefficients for such a function are given by

$$\begin{aligned}
 C_k &= \int_0^L g(x) e^{jk\omega_0 x} dx = \frac{1}{L} \left[ \int_0^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx + \int_{\frac{L}{2}}^L g(x) e^{jk\omega_0 x} dx \right] \\
 &= \frac{1}{L} \left[ \int_0^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx + \int_0^{\frac{L}{2}} g\left(u + \frac{L}{2}\right) e^{jk\omega_0(u + \frac{L}{2})} du \right] \\
 &= \frac{1}{L} \left[ \int_0^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx - \int_0^{\frac{L}{2}} g(u) e^{jk\omega_0 u} e^{jk\pi} du \right] \\
 C_k &= \begin{cases} 0 & \text{when } k \text{ is even} \\ \frac{2}{L} \int_0^{\frac{L}{2}} g(x) e^{jk\omega_0 x} dx & \text{when } k \text{ is odd} \end{cases}
 \end{aligned} \tag{5}$$

Thus a function with half-wave symmetry comprises of only odd-harmonics.